

Lecture 24. Eigenvalue Problems

Eigenvalue problems are particularly interesting in scientific computing, because the best algorithms for finding eigenvalues are powerful, yet particularly far from obvious. Here, we review the mathematics of eigenvalues and eigenvectors. Algorithms are discussed in later lectures.

Eigenvalues and Eigenvectors

Let $A \in \mathbb{C}^{m \times m}$ be a square matrix. A nonzero vector $x \in \mathbb{C}^m$ is an *eigenvector* of A , and $\lambda \in \mathbb{C}$ is its corresponding *eigenvalue*, if

$$Ax = \lambda x. \quad (24.1)$$

The idea here is that the action of a matrix A on a subspace S of $\mathbb{C}^m$ may sometimes mimic scalar multiplication. When this happens, the special subspace S is called an *eigenspace*, and any nonzero $x \in S$ is an eigenvector.

The set of all the eigenvalues of a matrix A is the *spectrum* of A , a subset of $\mathbb{C}$ denoted by $\Lambda(A)$.

Eigenvalue problems have a very different character from the problems involving square or rectangular linear systems of equations discussed in the previous lectures. For a system of equations, the domain of A could be one space and the range could be a different one. In Example 1.1, for example, A mapped n -vectors of polynomial coefficients to m -vectors of sampled polynomial values. To ask about the eigenvalues of such an A would be meaningless. Eigenvalue problems make sense only when the range and the domain spaces

are the same. This reflects the fact that in applications, eigenvalues are generally used where a matrix is to be compounded iteratively, either explicitly as a power A^k or implicitly in a functional form such as e^{tA} .

Broadly speaking, eigenvalues and eigenvectors are useful for two reasons, one algorithmic, the other physical. Algorithmically, eigenvalue analysis can simplify solutions of certain problems by reducing a coupled system to a collection of scalar problems. Physically, eigenvalue analysis can give insight into the behavior of evolving systems governed by linear equations. The most familiar examples in this latter class are the study of *resonance* (e.g., of musical instruments when struck or plucked or bowed) and of *stability* (e.g., of fluid flows subjected to small perturbations). In such cases eigenvalues tend to be particularly useful for analyzing behavior for large times t . See Exercise 24.3.

Eigenvalue Decomposition

An *eigenvalue decomposition* of a square matrix A , already mentioned in (5.1), is a factorization

$$A = X\Lambda X^{-1}. \quad (24.2)$$

(As we discuss below, such a factorization does not always exist.) Here X is nonsingular and Λ is diagonal.

This definition can be rewritten

$$AX = X\Lambda, \quad (24.3)$$

that is,

$$\left[\begin{array}{c} \\ \\ A \\ \\ \end{array} \right] \left[\begin{array}{c|c|c|c} x_1 & x_2 & \cdots & x_m \end{array} \right] = \left[\begin{array}{c|c|c|c} x_1 & x_2 & \cdots & x_m \end{array} \right] \left[\begin{array}{cccc} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_m \end{array} \right].$$

This makes it clear that if x_j is the j th column of X and λ_j is the j th diagonal entry of Λ , then $Ax_j = \lambda_j x_j$. Thus the j th column of X is an eigenvector of A and the j th entry of Λ is the corresponding eigenvalue.

The eigenvalue decomposition expresses a change of basis to “eigenvector coordinates.” If $Ax = b$ and $A = X\Lambda X^{-1}$, we have

$$(X^{-1}b) = \Lambda(X^{-1}x). \quad (24.4)$$

Thus, to compute Ax , we can expand x in the basis of columns of X , apply Λ , and interpret the result as a vector of coefficients of a linear combination of the columns of X .

Geometric Multiplicity

As stated above, the set of eigenvectors corresponding to a single eigenvalue, together with the zero vector, forms a subspace of $\mathbb{C}^m$ known as an *eigenspace*. If λ is an eigenvalue of A , let us denote the corresponding eigenspace by E_λ . An eigenspace E_λ is an example of an *invariant subspace* of A ; that is, $AE_\lambda \subseteq E_\lambda$.

The dimension of E_λ can be interpreted as the maximum number of linearly independent eigenvectors that can be found, all with the same eigenvalue λ . This number is known as the *geometric multiplicity* of λ . The geometric multiplicity can also be described as the dimension of the nullspace of $A - \lambda I$, since that nullspace is again E_λ .

Characteristic Polynomial

The *characteristic polynomial* of $A \in \mathbb{C}^{m \times m}$, denoted by p_A or simply p , is the degree m polynomial defined by

$$p_A(z) = \det(zI - A). \quad (24.5)$$

Thanks to the placement of the minus sign, p is *monic*: the coefficient of its degree m term is 1.

Theorem 24.1. λ is an eigenvalue of A if and only if $p_A(\lambda) = 0$.

Proof. This follows from the definition of an eigenvalue:

$$\begin{aligned} \lambda \text{ is an eigenvalue} &\iff \text{there is a nonzero vector } x \text{ such that } \lambda x - Ax = 0 \\ &\iff \lambda I - A \text{ is singular} \\ &\iff \det(\lambda I - A) = 0. \end{aligned} \quad \square$$

Theorem 24.1 has an important consequence. *Even if a matrix is real, some of its eigenvalues may be complex.* Physically, this is related to the phenomenon that real dynamical systems can have motions that oscillate as well as grow or decay. Algorithmically, it means that even if the input to a matrix eigenvalue problem is real, the output may have to be complex.

Algebraic Multiplicity

By the fundamental theorem of algebra, we can write p_A in the form

$$p_A(z) = (z - \lambda_1)(z - \lambda_2) \cdots (z - \lambda_m) \quad (24.6)$$

for some numbers $\lambda_j \in \mathbb{C}$. By Theorem 24.1, each λ_j is an eigenvalue of A , and all eigenvalues of A appear somewhere in this list. In general, an eigenvalue might appear more than once. We define the *algebraic multiplicity*

of an eigenvalue λ of A to be its multiplicity as a root of p_A . An eigenvalue is *simple* if its algebraic multiplicity is 1.

The characteristic polynomial gives us an easy way to count the number of eigenvalues of a matrix.

Theorem 24.2. *If $A \in \mathbb{C}^{m \times m}$, then A has m eigenvalues, counted with algebraic multiplicity. In particular, if the roots of p_A are simple, then A has m distinct eigenvalues.*

Note that in particular, every matrix has at least one eigenvalue.

The algebraic multiplicity of an eigenvalue is always at least as great as its geometric multiplicity. To prove this, we need to know something about similarity transformations.

Similarity Transformations

If $X \in \mathbb{C}^{m \times m}$ is nonsingular, then the map $A \mapsto X^{-1}AX$ is called a *similarity transformation* of A . We say that two matrices A and B are *similar* if there is a similarity transformation relating one to the other, i.e., if there exists a nonsingular $X \in \mathbb{C}^{m \times m}$ such that $B = X^{-1}AX$. As described above in the special case of the diagonalization (24.2), any similarity transformation is a change of basis operation.

Many properties are shared by similar matrices A and $X^{-1}AX$.

Theorem 24.3. *If X is nonsingular, then A and $X^{-1}AX$ have the same characteristic polynomial, eigenvalues, and algebraic and geometric multiplicities.*

Proof. The proof that the characteristic polynomials match is a straightforward computation:

$$\begin{aligned} p_{X^{-1}AX}(z) &= \det(zI - X^{-1}AX) = \det(X^{-1}(zI - A)X) \\ &= \det(X^{-1}) \det(zI - A) \det(X) = \det(zI - A) = p_A(z). \end{aligned}$$

From the agreement of the characteristic polynomials, the agreement of the eigenvalues and algebraic multiplicities follows. Finally, to prove that the geometric multiplicities agree, we can verify that if E_λ is an eigenspace for A , then $X^{-1}E_\lambda$ is an eigenspace for $X^{-1}AX$, and conversely. $\square$

We can now relate geometric multiplicity to algebraic multiplicity.

Theorem 24.4. *The algebraic multiplicity of an eigenvalue λ is at least as great as its geometric multiplicity.*

Proof. Let n be the geometric multiplicity of λ for the matrix A . Form an $m \times n$ matrix $\hat{V}$ whose n columns constitute an orthonormal basis of the eigenspace $\{x : Ax = \lambda x\}$. Then, extending $\hat{V}$ to a square unitary matrix V , we obtain V^*AV in the form

$$B = V^*AV = \begin{bmatrix} \lambda I & C \\ 0 & D \end{bmatrix}, \quad (24.7)$$

where I is the $n \times n$ identity, C is $n \times (m-n)$, and D is $(m-n) \times (m-n)$. By the definition of the determinant, $\det(zI - B) = \det(zI - \lambda I) \det(zI - D) = (z - \lambda)^n \det(zI - D)$. Therefore the algebraic multiplicity of λ as an eigenvalue of B is at least n . Since similarity transformations preserve multiplicities, the same is true for A . $\square$

Defective Eigenvalues and Matrices

Although a generic matrix has algebraic and geometric multiplicities that are equal (namely, all 1), this is by no means true of every matrix.

Example 24.1. Consider the matrices

$$A = \begin{bmatrix} 2 & & \\ & 2 & \\ & & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 & \\ & 2 & 1 \\ & & 2 \end{bmatrix}.$$

Both A and B have characteristic polynomial $(z - 2)^3$, so there is a single eigenvalue $\lambda = 2$ of algebraic multiplicity 3. In the case of A , we can choose three independent eigenvectors, for example e_1 , e_2 , and e_3 , so the geometric multiplicity is also 3. For B , on the other hand, we can find only a single independent eigenvector (a scalar multiple of e_1), so the geometric multiplicity of the eigenvalue is only 1. $\square$

An eigenvalue whose algebraic multiplicity exceeds its geometric multiplicity is a *defective eigenvalue*. A matrix that has one or more defective eigenvalues is a *defective matrix*.

Any diagonal matrix is nondefective. For such a matrix, both the algebraic and the geometric multiplicities of an eigenvalue λ are equal to the number of its occurrences along the diagonal.

Diagonalizability

The class of nondefective matrices is precisely the class of matrices that have an eigenvalue decomposition (24.2).

Theorem 24.5. *An $m \times m$ matrix A is nondefective if and only if it has an eigenvalue decomposition $A = X\Lambda X^{-1}$.*

Proof. ($\Leftarrow$) Given an eigenvalue decomposition $A = X\Lambda X^{-1}$, we know by Theorem 24.3 that Λ is similar to A , with the same eigenvalues and the same multiplicities. Since Λ is a diagonal matrix, it is nondefective, and thus the same holds for A .

($\Rightarrow$) A nondefective matrix must have m linearly independent eigenvectors, because eigenvectors with different eigenvalues must be linearly independent, and each eigenvalue can contribute as many linearly independent eigenvectors as its multiplicity. If these m independent eigenvectors are formed into the columns of a matrix X , then X is nonsingular and we have $A = X\Lambda X^{-1}$. $\square$

In view of this result, another term for nondefective is *diagonalizable*.

Does a diagonalizable matrix A in some sense “behave like” its diagonal equivalent Λ ? The answer depends on what aspect of behavior one measures and on the condition number of X , the matrix of eigenvectors. If X is highly ill-conditioned, then a great deal of information may be discarded in passing from A to Λ . See “A Note of Caution: Nonnormality” in Lecture 34.

Determinant and Trace

The *trace* of $A \in \mathbb{C}^{m \times m}$ is the sum of its diagonal elements: $\text{tr}(A) = \sum_{j=1}^m a_{jj}$. Both the trace and the determinant are related simply to the eigenvalues.

Theorem 24.6. *The determinant $\det(A)$ and trace $\text{tr}(A)$ are equal to the product and the sum of the eigenvalues of A , respectively, counted with algebraic multiplicity:*

$$\det(A) = \prod_{j=1}^m \lambda_j, \quad \text{tr}(A) = \sum_{j=1}^m \lambda_j. \quad (24.8)$$

Proof. From (24.5) and (24.6), we compute

$$\det(A) = (-1)^m \det(-A) = (-1)^m p_A(0) = \prod_{j=1}^m \lambda_j.$$

This establishes the first formula. As for the second, from (24.5), it follows that the coefficient of the z^{m-1} term of p_A is the negative of the sum of the diagonal elements of A , or $-\text{tr}(A)$. On the other hand, from (24.6), this coefficient is also equal to $-\sum_{j=1}^m \lambda_j$. Thus $\text{tr}(A) = \sum_{j=1}^m \lambda_j$. $\square$

Unitary Diagonalization

It sometimes happens that not only does an $m \times m$ matrix A have m linearly independent eigenvectors, but these can be chosen to be orthogonal. In such

a case, A is *unitarily diagonalizable*, that is, there exists a unitary matrix Q such that

$$A = Q\Lambda Q^*. \quad (24.9)$$

This factorization is both an eigenvalue decomposition and a singular value decomposition, aside from the matter of the signs (possibly complex) of the entries of Λ .

We have already seen a class of matrices that are unitarily diagonalizable: the hermitian matrices. The following result follows from Theorem 24.9, below.

Theorem 24.7. *A hermitian matrix is unitarily diagonalizable, and its eigenvalues are real.*

The hermitian matrices are not the only ones that are unitarily diagonalizable. Other examples include skew-hermitian matrices, unitary matrices, circulant matrices, and any of these plus a multiple of the identity. In general, the class of matrices that are unitarily diagonalizable have an elegant characterization. By definition, we say that a matrix A is *normal* if $A^*A = AA^*$. The following result is well known.

Theorem 24.8. *A matrix is unitarily diagonalizable if and only if it is normal.*

Schur Factorization

One final matrix factorization is actually the one that is most useful in numerical analysis, because all matrices, including defective ones, can be factored in this way. A *Schur factorization* of a matrix A is a factorization

$$A = QTQ^*, \quad (24.10)$$

where Q is unitary and T is upper-triangular. Note that since A and T are similar, the eigenvalues of A necessarily appear on the diagonal of T .

Theorem 24.9. *Every square matrix A has a Schur factorization.*

Proof. We proceed by induction on the dimension m of A . The case $m = 1$ is trivial, so suppose $m \geq 2$. Let x be any eigenvector of A , with corresponding eigenvalue λ . Take x to be normalized and let it be the first column of a unitary matrix U . Then, just as in (24.7), it is easily checked that the product U^*AU has the form

$$U^*AU = \begin{bmatrix} \lambda & B \\ 0 & C \end{bmatrix}.$$

By the inductive hypothesis, there exists a Schur factorization VTV^* of C . Now write

$$Q = U \begin{bmatrix} 1 & 0 \\ 0 & V \end{bmatrix}.$$

This is a unitary matrix, and we have

$$Q^*AQ = \begin{bmatrix} \lambda & BV \\ 0 & T \end{bmatrix}.$$

This is the Schur factorization we seek. $\square$

Eigenvalue-Revealing Factorizations

In the preceding pages we have described three examples of *eigenvalue-revealing factorizations*, factorizations of a matrix that reduce it to a form in which the eigenvalues are explicitly displayed. We can summarize these as follows.

A diagonalization $A = X\Lambda X^{-1}$ exists if and only if A is nondefective.

A unitary diagonalization $A = Q\Lambda Q^*$ exists if and only if A is normal.

A unitary triangularization (Schur factorization) $A = QTQ^*$ always exists.

To compute eigenvalues, we shall construct one of these factorizations. In general, this will be the Schur factorization, since this applies without restriction to all matrices. Moreover, since unitary transformations are involved, the algorithms that result tend to be numerically stable. If A is normal, then the Schur form comes out diagonal, and in particular, if A is hermitian, then we can take advantage of this symmetry throughout the computation and reduce A to diagonal form with half as much work or less than is required for general A .

Exercises

24.1. For each of the following statements, prove that it is true or give an example to show it is false. Throughout, $A \in \mathbb{C}^{m \times m}$ unless otherwise indicated, and “ew” stands for eigenvalue. (This comes from the German “Eigenwert.” The corresponding abbreviation for eigenvector is “ev,” from “Eigenvektor.”)

- (a) If λ is an ew of A and $\mu \in \mathbb{C}$, then $\lambda - \mu$ is an ew of $A - \mu I$.
- (b) If A is real and λ is an ew of A , then so is $-\lambda$.
- (c) If A is real and λ is an ew of A , then so is $\bar{\lambda}$.
- (d) If λ is an ew of A and A is nonsingular, then λ^{-1} is an ew of A^{-1} .
- (e) If all the ew's of A are zero, then $A = 0$.
- (f) If A is hermitian and λ is an ew of A , then $|\lambda|$ is a singular value of A .
- (g) If A is diagonalizable and all its ew's are equal, then A is diagonal.